KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY
COLLEGE OF ENGINEERING
DEPARTMENT OF AGRICULTURAL ENGINEERING

AE 352 SOIL MECHANICS APPLICATIONS TO MECHANIZATION

TUTORIAL QUESTIONS

1. The size of a particle in a soil deposit has a fundamental influence on the properties and engineering behaviour of the deposit. (*True or False*?).

Answer ***questions 2 to 9*** by completing the following soil classification table:

Particle description	**Particle Size (mm)**			
	British Standard (BS)	**AASHTO**	**ASTM**	**Unified**
Gravel	60 – 2	**Q 4**	>2	**Q 8**
Sand	**Q 2**	**Q 5**	**Q 6**	**Q 9**
Silt	**Q 3**	0.05 – 0.002	**Q 7**	<0.075
Clay	<0.002	<0.002	<0.005	

AASHTO – American Association of State Highways and Transportation Officials; ASTM – American Society for Testing of Materials

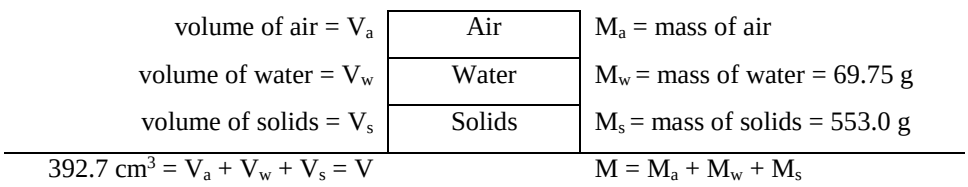
10. Name *two* forces that influence the engineering behaviour of soils:
    a)
    b)

11. List *two* categories of engineering problems encountered in the context of investigating the stresses and deformation in a soil mass.
    a)
    b)

12. If a soil is unsaturated, the pores are said to be filled with:
    a) water
    b) solids
    c) water and air
    d) air

13. A soil is said to be not only porous but permeable and this characteristic of permeability fundamentally influences the engineering behaviour of a soil deposit in many practical situations. (*True or False?*).

The diagram below shows the proportion by mass and volume of air, water and solids in the soil. Also, volume of voids, $V_v = V_a + V_w$. Soil particle density is 2.65 Mg/m^3.



Refer to the figure above and answer the following 9 questions.

14. What is the mass water content of the soil?
    a) 0.178
    b) 0.126
    c) 0.700
    d) None of the above

15. What is the volumetric water content of the soil?
    a) 0.178
    b) 0.126
    c) 0.700
    d) None of the above

16. Calculate the wet bulk density of the soil.
    a) 1,586 kg/m^3
    b) 1,877 kg/m^3
    c) 1,408 kg/m^3
    d) None of the above

17. Calculate the dry bulk density of the soil.
    a) 1,586 kg/m^3
    b) 1,877 kg/m^3
    c) 1,408 kg/m^3
    d) None of the above

18. What is the saturated density of the soil?
    a) 1,586 kg/m^3
    b) 1,877 kg/m^3
    c) 1,408 kg/m^3
    d) None of the above

19. What is the porosity of the soil?
    a) 0.178
    b) 0.126
    c) 0.700
    d) 0.469

20. Determine the void ratio of the soil.
    a) 0.882
    b) 0.469
    c) 0.700
    d) None of the above

21. Determine the degree of saturation of the soil.
    a) 0.279
    b) 0.469
    c) 0.700
    d) None of the above

22. What is the air content of the soil?
    a) 0.379
    b) 0.469
    c) 0.291
    d) None of the above

23. Which of the following primary objectives of soil *compaction* is *not* true?
    a) Increasing the shear strength and thereby improving the stability of embankments and the bearing capacity of foundations and pavements
    b) Increasing soil compressibility and thereby reducing settlements
    c) Decreasing void ratio and thereby reducing soil permeability
    d) Reducing the potential for swelling and shrinkage

24. The energy expended during the *compaction* process is called:
    a) Compressive load
    b) Shear force
    c) Compactive effort
    d) Bearing capacity

For questions 25 to 28, state whether the statements are true or false regarding soil compaction:

25. It is not possible to remove water from the voids by compaction but the addition of water to a slightly moist soil facilitates compaction by reducing *surface tension*.

26. Maximum dry density is achieved at optimum water content beyond which the density declines.

27. A higher compactive effort gives a higher maximum dry density but lower optimum water content for the same soil.

28. Relative compaction under field conditions should range between 90 and 100 % as satisfactory requirement for compaction works.

Problem: In constructing a farm road with clay fill, the embankment is compacted to a bulk density of 2.05 Mg/m^3 at water content of 18 % by mass. The specific gravity of the soil particles is 2.70.

Answer the next 4 questions from the problem stated above.

29. Calculate the porosity of the road embankment.
    a) 0.257
    b) 0.357
    c) 0.457
    d) None of the above

30. Determine the degree of saturation of the road embankment.
    a) 0.313
    b) 0.357

c) 0.877
d) None of the above

31. What will the air content of the road embankment be?
a) 0.313
b) 0.344
c) 0.044
d) 0.257

32. The dry bulk density of the road embankment at the end of the construction will be:
a) 1.737 Mg/m^3
b) 1.357 Mg/m^3
c) 1.877 Mg/m^3
d) none on the above

33. Draw a labelled stress-strain diagram for an *ideal* soil.

34. The stress distribution that results in a soil mass due to loads applied by construction of engineering works are influenced by the stress-strain properties of the soil. (*True* or *False*?)

35. Fill in the blanks for the following statement: Soils are said to be very *weak* in __(a)__, very *strong* in __(b)__ and *fail* only by __(c).__

36. If a coarse-grained soil contains approximately equal proportion of all particle sizes and is characterised by a relatively smooth grading curve, it is described as:
a) uniformly-graded
b) well-graded
c) gap-graded
d) None of the above

37. A soil sample that has a high proportion of the particle sizes lying in a narrow size band is graded as:
a) uniform
b) well-graded
c) gap-graded
d) None of the above

38. A coarse-grained soil that exhibits an absence of particles in the intermediate size range is considered as:
a) uniformly-graded
b) well-graded
c) gap-graded
d) None of the above

39. An indication of the grading of a soil may be expressed numerically in terms of the coefficient of uniformity (C_u) which is expressed as:
a) D_{30}/D_{10}
b) $D_{30}^2/[D_{10}.D_{60}]$
c) D_{60}/D_{10}
d) None of the above

40. Which of the following describes a coarse-grained soil that contains approximately equal proportions of all particle sizes? (C_z = coefficient of curvature).

a) $1<C_z<3$
b) $C_z>3$
c) $4>C_z>6$
d) None of the above

41. A uniformly-graded coarse-grained soil could be described as poorly graded (*True or False?*).

42. Atterberg limits are simply

a) soil shear strengths.
b) soil water contents.
c) soil plasticity.
d) None of the above

43. Which of the following is not a consistency state of clay soil?

a) Plastic state
b) Non-plastic state
c) Semi-plastic state
d) Liquid state

44. The range of water contents over which a clay soil exhibit plasticity is termed

a) Plasticity index
b) Liquidity index
c) Atterberg limit
d) Activity

45. The natural water contents at which a clay soil existing in a field may be compared with its plastic and liquid limits is known as

a) Plasticity index
b) Liquidity index
c) Atterberg limit
d) Activity

46. Which of the following is *not* a clay soil?

a) kaolinite
b) illite
c) montmorillonite
d) quartzite

47. Which of the following statements about the activity of clay is *not* true?

a) activity of kaolinite is less than that of montmorillonite
b) illite has higher activity than kaolinite
c) montmorillonite has higher activity than illite
d) illite has same activity as quartzite

48. Plasticity of clay soil is attributed to the deformation of adsorbed water around clay mineral particles. This phenomenon is known as:

a) Plasticity index
b) Liquidity index
c) Activity
d) None of the above

49. If the water content of a clay soil in a field equals its *liquid limit*, what will be the value of its *plasticity index*?
a) 0
b) 0.5
c) 1
d) None of the above

50. If the water content of a clay soil in a field equals its *plastic limit*, what will be the value of its *liquidity index*?
a) 0
b) 0.5
c) 1
d) None of the above

51. If the water content of a clay soil in a field equals its *liquid limit*, what will be the value of its *liquidity index*?
a) 0
b) 0.5
c) 1
d) None of the above

52. If the liquidity index of clay soil is equal to 1.0, its field water content will be equal to the sum of its *plasticity index* and *plastic limit*. (*True or False*?)

53. The *shear strength* of a soil is.....

54. Shear strength determines such factors as:
a) the allowable bearing capacity of a foundation
b) the activity of a clay soil
c) particle size distribution of a soil
d) the Atterberg limits of a clay soil

55. Since water cannot sustain shear stress, the shear resistance of the soil must result solely from the frictional resistance acting at the particle contact point (*True* or *False*?).

56. State whether the following equation is *true* or *false*:

$$\rho_1 = \rho_3 N_\varphi + 2C\sqrt{N_\varphi}$$

where, the symbols have their usual meanings.

Problem: A series of *undrained tri-axial* tests on samples of saturated soil gave the following result:

Lateral pressure (kN/m^2)	100	200	300
Pore water pressure (kN/m^2)	20	70	136
Principal stress difference at failure (kN/m^2)	290	400	534

Answer the next 4 questions from the data provided above. Show the solution on a graph sheet and attach to the answer sheet.

57. Determine the soil cohesion with respect to *total stress*.
    e) 12 kN/m^2
    f) 60 kN/m^2
    g) 36 kN/m^2
    h) None of the above

58. What is the value of soil cohesion with respect to *effective stress*?
    a) 12 kN/m^2
    b) 60 kN/m^2
    c) 36 kN/m^2
    d) None of the above

59. What should the angle of internal soil friction be with regard to *total stress*?
    a) 12^0
    b) 22^0
    c) 36^0
    d) None of the above

60. Compute the angle of internal soil friction with regard to *effective stress*.
    a) 12^0
    b) 22^0
    c) 36^0
    d) None of the above

61. In order to achieve an efficient tillage operation using an animal-drawn implement, there is the need achieve a good interaction among the following:
    a) Man-implement-soil
    b) Animal-implement-soil
    c) Man-animal-implement-soil
    d) None of the above

62. In order to achieve an efficient tillage operation using a machine-drawn implement, there is the need achieve a good interaction among the following:
    a) Machine-implement-soil
    b) Man-machine-implement-soil
    c) Man-implement-soil
    d) None of the above

63. Which of the following factors does *not* affect draught force in a machine-drawn tillage operation?
    a) Soil-implement interaction.
    b) Forward speed of the machine.
    c) Rake angle of the implement.

d) None of the above.

64. What upper limit of rake angles is recommended to keep the draught force low in a tillage operation?
a) 10^0
b) 50^0
c) 90^0
d) None of the above

65. Make a *labelled* sketch showing the relationship between rake angle and draught force.

66. What upper limit of rake angles is recommended to keep the draught force low in a tillage operation?
a) 10^0
b) 50^0
c) 90^0
d) None of the above

67. For wide tines, at what rake angle does a curved blade behave as a straight one irrespective of its curvature?
a) 50^0
b) 90^0
c) 105^0
d) None of the above

68. Using the rupture-distance ratio charts, calculate the rupture distance of the soil for a soil which has no adhesion between it and the implement of 60^0 rake angle and 25 cm depth of cut. The angle of internal soil friction is 45^0.
a) 0.03 m
b) 0.75 m
c) 1.00 m
d) None of the above

69. List the parts of a conventional *subsoiler*

70. Make *well-labelled* sketches of the *cross-section* of soil disturbance for the following subsoiler arrangements:
a) Winged subsoiler with leading shallow tines
b) Winged subsoiler alone
c) Conventional subsoiler

71. The attachment of leading shallow tines to the subsoiler frame assists in reducing the strength of the undisturbed surface layers before subsoiling and results in increased subsoil disturbance. (*True or False?).*

72. A ploughing operation followed by a subsoiling one results in increased subsoil disturbance. (*True or False*?)

73. Under which of the following field conditions is *subsoiling* not useful for agriculture?
a) Soils with drainage problems caused by slow soil permeability

b) Deep, well-drained soils with good soil structure
c) Soils with subsoil hard pans caused by farm machinery
d) Soils with natural hard pans

74. A typical draught force required for the operation of a *conventional* subsoiler at a soil depth of about 42 cm is in the range of about:
a) 44 kN
b) 40 kN
c) 27 kN
d) 21 kN

75. A typical draught force required for the operation of a *winged* subsoiler at a soil depth of about 42 cm is in the range of about:
a) 44 kN
b) 40 kN
c) 27 kN
d) 21 kN

76. The shattering effect of *subsoiling* on a compacted layer of soil creates extensive cracks to the soil surface, thus producing some effects on the soil. Which is the odd one out of the following?
a) Soil aeration is improved
b) Soil drainage is improved
c) Deeper rooting is permitted
d) None of the above

77. What is the approximate *spacing-working depth ratio* needed to achieve nearly level soil surface for the implement arrangements of a *winged subsoiler with leading shallow tines*?
a) 1.50
b) 2.00
c) 2.25
d) None of the above

78. What is the approximate *spacing-working depth ratio* needed to achieve nearly level soil surface for the implement arrangements of a *winged subsoiler* alone?
a) 1.50
b) 2.00
c) 2.25
d) None of the above

79. What is the approximate *spacing-working depth ratio* needed to achieve nearly level soil surface for the implement arrangements of a *conventional subsoiler*?
a) 1.50
b) 2.00
c) 2.25
d) None of the above

80. A vertical bulldozer blade 2 m wide cuts through a soil 100 mm deep. The soil has the following properties: bulk unit density = 15 kN/m^3; soil cohesion = 10 kN/m^2; angle of internal soil friction = 30^0; angle of soil-blade friction = 20^0. The horizontal soil surface level

with the top of the blade is surcharged with a uniform depth of soil at 12 kN/m^2. Determine the resultant force required to pull the blade. Use the N charts.

a) 12.66 kN
b) 11.21 kN
c) 24.70 kN
d) None of the above

81. Results of a sieve analysis of a soil were as follows:

Sieve Size	Wt. of soil retained (g)
50 mm	0
37.5 mm	15.5
19 mm	17
12.5 mm	10
8 mm	11
5.9 mm	33
4.75 mm	33.5
2.8 mm	81
2360 µm	18
1300 µm	31
400 µm	32.5
212 µm	9
150 µm	8
100 µm	5.5
75 µm	5

The total weight of the sample was 311 g. Plot the particle size distribution curve and give a description of the soil. Also determine the C_u and C_z.

82. The following results were obtained from a liquid limit test on a fine-grained soil:

Penetration (mm)	15.6	18.2	21.4	23.6
MC (%)	48.6	54.8	62.2	67.4

A plastic limit test gave a value of 22 %. Determine the liquid limit and plasticity index of the soil.

83. Assuming there is no adhesion between a tine blade and the moving soil, compute the total force acting on a 12 cm wide tine. Specific gravity of soil = 1.5; angle of soil-soil friction = 30°; depth of cut of implement = 25 cm; soil cohesion = 12 kN/m^2; soil surcharge = 3.5 kN/m^2 [Use the N factor charts]. If soil adhesion is 5 kN/m^2 and the angle of soil metal friction is 12.5^o, what will be the total force acting on the blade? Rake angle = 70°.

84. Standard Proctor compaction tests carried out on a sample of sandy clay yielded the following results:

Wet bulk density (kg/m^3)	2058	2125	2152	2159	2140
Water content (% mass)	12.9	14.3	15.7	16.9	17.9

a) Plot the curve of dry bulk density against water content and hence find the maximum dry bulk density and the optimum water content.

b) Calculate the water content necessary for complete saturation at this maximum dry bulk density if the specific gravity of the soil particles is 2.70.
c) Plot the *zero air voids* and 5 % air voids line superimposed on the same diagram.

85. An investigation at a particular site revealed a deposit of sand overlying bedrock. A sample of sand taken from above the water table had a total mass of 2205 g and was found to have occupied in its natural state a volume of 1125 cm^3. After oven-drying the mass of the sample was 1970 g and its particle specific gravity was found to be 2.65. a) For the sand above the water table, calculate the wet bulk density, water content by mass, void ratio, degree of saturation and air voids content. b) For the sand below the water table, calculate water content by mass and the submerged density of the soil.

86. A _____is a vehicle specifically designed to deliver a high tractive effort (or torque) at slow speeds, for the purposes of hauling a trailer or machinery used in agriculture or construction.

87. The part of a bulldozer that gives it better traction, and makes it more ideal for the ups and downs of rough terrain is called what?

88. *Draglines* and *power shovels* are the _____ forms of excavation equipment.
a) primary
b) secondary
c) tertiary
d) None of the above

89. Which of the following earth-moving equipment is used to excavate deep holes?
a) front-end loader
b) back-end loader
c) dragline
d) None of the above

90. Scrapers are
a) finishing equipment that smoothen earth material placed on the surface
b) used to level and contour land, as in road construction
c) used to dig holes
d) None of the above

91. Lightweight tractor equipped with a *backhoe*, which is an open scoop attached rigidly to a hinged boom, can be used
a) to dig shallow trenches
b) to level and contour land, as in road construction
c) to dig deep holes
d) None of the above

92. Which earth-moving machine is used to clear brush or debris, remove boulders, and level the ground?
a) front-end loader
b) dragline
c) bulldozer
d) None of the above

93. The removal of earth or material from the bottom of water bodies is performed by
    a) dragline
    b) dredges
    c) scraper
    d) None of the above

94. Power shovels are fitted with specially designed buckets called _____, which dig into the earth and shovel it up. Its bottom can then be opened to dump the soil into a truck for removal.
    a) clamshell
    b) loader
    c) scraper
    d) None of the above

95. A contractor intends laying pipes and cables underground. Which of the following earth-moving machines will you recommend?
    a) excavators and trenchers
    b) power shovels
    c) dragline
    d) None of the above

96. _____ are the towers that use cables, pulleys, and a scoop or hook to lift and lower a payload
    a) draglines
    b) dredges
    c) cranes
    d) None of the above

97. In rehabilitating a farm road, what equipment will you recommend for breaking up the existing pavement before re-construction?
    a) backhoe
    b) front hoe
    c) power shovel
    d) None of the above

98. A backhoe loader is a tractor with a loader in front of it, and a backhoe attachment in the rear (*True* or *False*?).

99. After how many hours of operation would you recommend the replacement of the bucket teeth and rubber track for a compact excavator?
    a) 20 hours
    b) 200 hours
    c) 2000 hours
    d) None of the above

100. Newer versions of bulldozers can scoop and load earth material in a combined operation. Moreover, they can hold the ground well, and use a torque divider between their 2 tracks to facilitate more or less dragging power (*True* or *False*?)